

Does online truncated backpropagation work sometimes?

Kernel Limit of Recurrent Neural Networks Trained on Ergodic Data Sequences

Samuel Lam¹, Justin Sirignano¹, Konstantinos Spiliopoulos²

¹Mathematical Institute, University of Oxford, ² Department of Mathematics & Statistics, Boston University
{samuel.lam, justin.sirignano}@maths.ox.ac.uk, kspiliop@bu.edu



UNIVERSITY OF
OXFORD

Mathematical
Institute

Overview

1. A **law of large number** for infinite-wide RNN.
2. The average RNN memories converge to a **random fixed point**.
3. A **gradient flow** for truncated backpropagation through time, in which a **kernel** emerges.

The gradient flow: Let $\varsigma_{x,u}(\cdot) = \sigma(\cdot^\top x + u)$. As $N \rightarrow \infty$, we have $g_{[Nt]}^N(h) \approx g_t(h)$, and

$$\frac{d}{dt} \left[\frac{1}{2} \int (g_t(\varsigma_{\tilde{x},\tilde{u}}) - \tilde{y})^2 \mu(d\tilde{H}) \right] = -\alpha \iint \mathcal{K}_\lambda(\varsigma_{\tilde{x},\tilde{u}}, \varsigma_{x,u}) (g_t(\varsigma_{\tilde{x},\tilde{u}}) - \tilde{y}) (g_t(\varsigma_{x,u}) - y) \mu(dH) \mu(d\tilde{H}) \leq 0.$$

$$\mathcal{K}_\lambda(\varsigma_{x,u}, \varsigma_{\tilde{x},\tilde{u}}) = \int [\sigma(w^\top x + u) \sigma(w^\top \tilde{x} + \tilde{u}) + \mu_{e^2} \sigma'(w^\top x + u) \sigma'(w^\top \tilde{x} + \tilde{u}) x^\top \tilde{x}] \lambda(dw).$$

Settings

- Input $(X_k)_{k \geq 0}$ with latent states $(Z_k)_{k \geq 0}$, output $(Y_k)_{k \geq 0}$ that depends on the i.i.d. noise $(\eta_k)_{k \geq 0}$:

$$(X_{k+1}, Z_{k+1}) \mid (X_k, Z_k) \sim P((X_k, Z_k), \cdot),$$

$$Y_k = f(X_k, Z_k, \eta_k).$$

The transition probability P is a **contraction**.

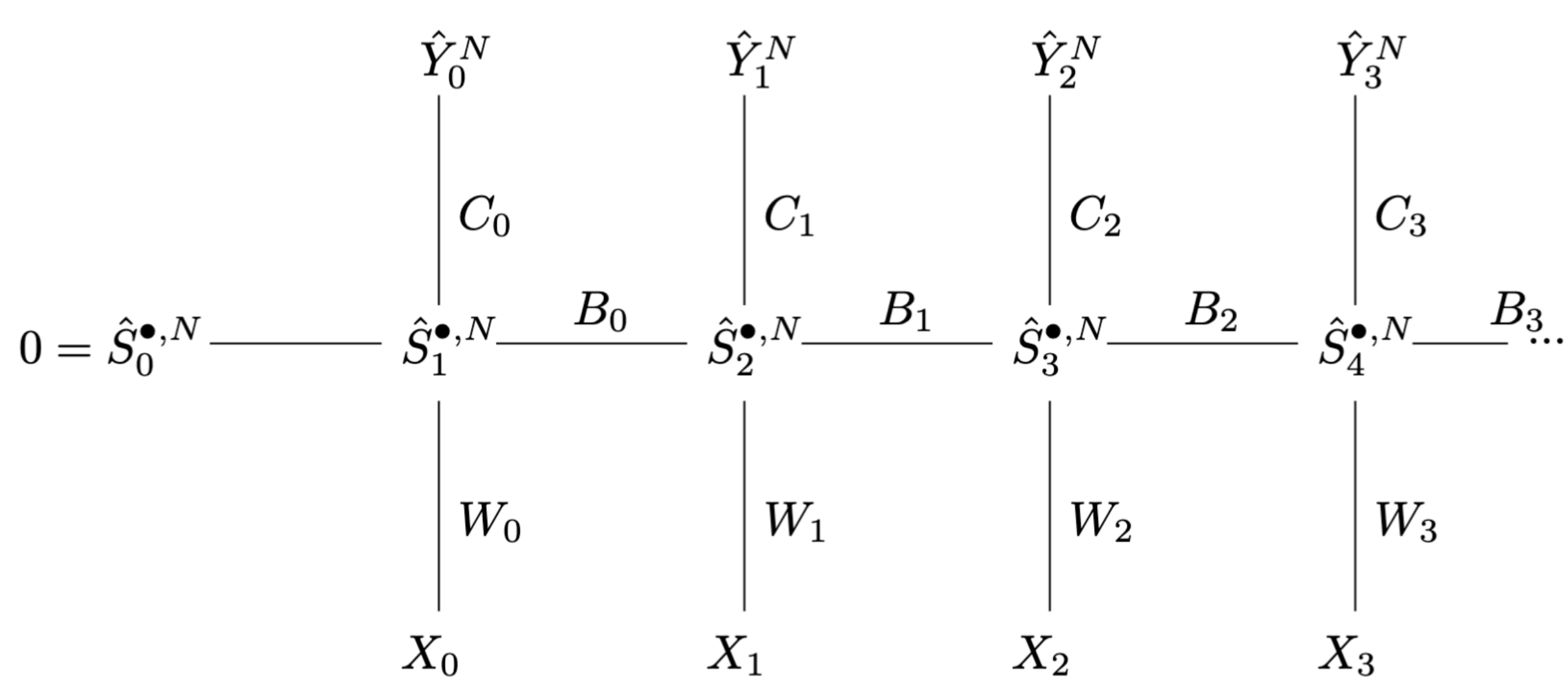
- Standard (Elman) RNN: (here $\beta \in (0.5, 1)$)

$$S_{k+1}^{i,N} := \sigma((W^i)^\top X_k + u_k^N), \quad S_0^{i,N} = 0,$$

$$u_k^N = \frac{1}{N} \sum_{j=1}^N B^j S_k^{j,N}, \quad g_k^N(X) := \frac{1}{N\beta} \sum_{i=1}^N C^i S_{k+1}^{i,N}.$$

σ is twice-differentiable with bounded outputs and derivatives.

- Trained by (1-step) **online truncated back-propagation**. [1]



$$\theta_{k+1} = \theta_k - \frac{\alpha}{N^{2-2\beta}} \nabla_{\theta} \ell_k$$

- Parameters are tested against test functions:

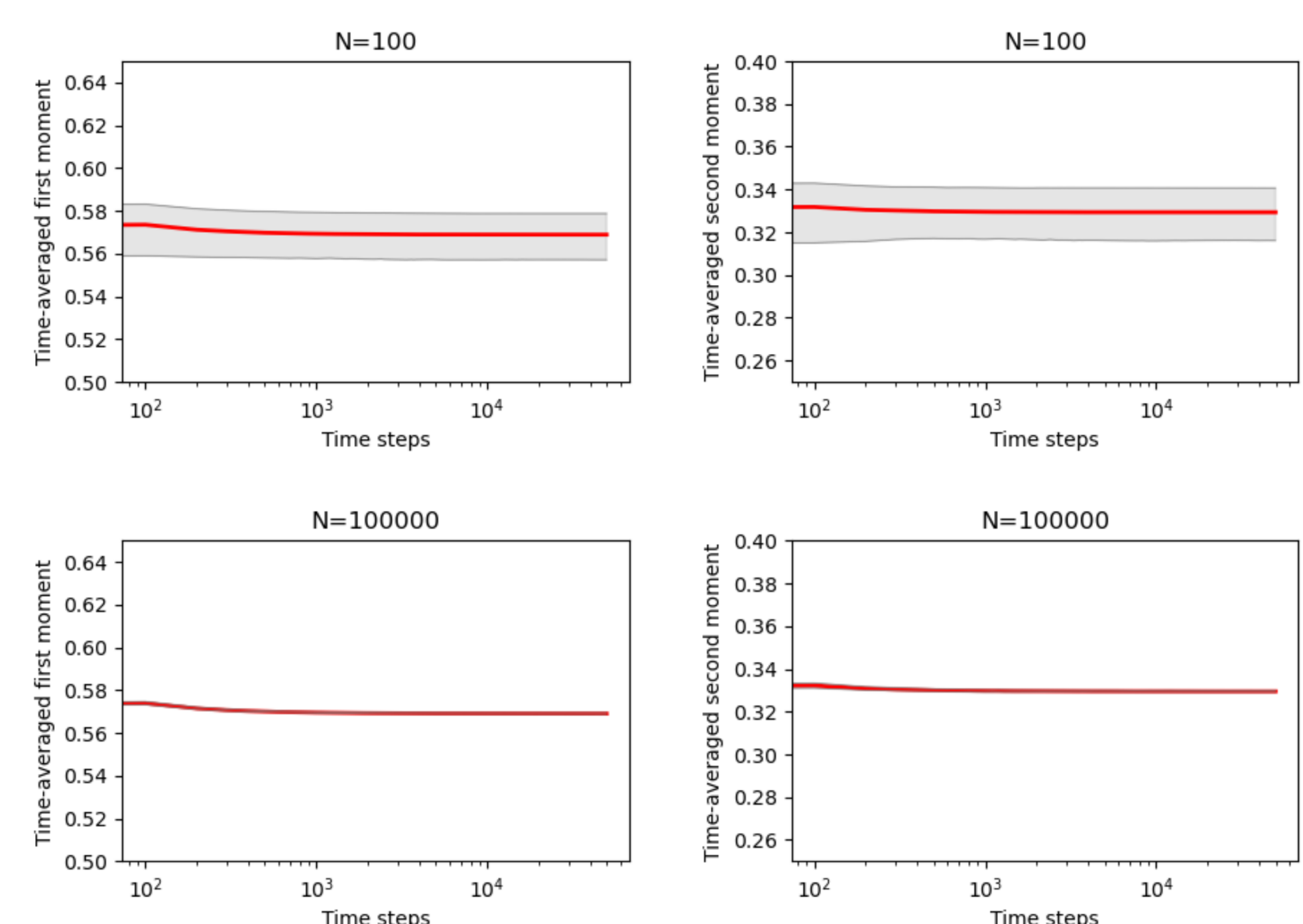
$$g_k^N(h) = \frac{1}{N\beta} \sum_{i=1}^N C_k^i h(W_k^i).$$

Law of Large Numbers + Random Fixed Point

Law of Large Numbers: the average memories u_k^N converges in L^2 to the following process:

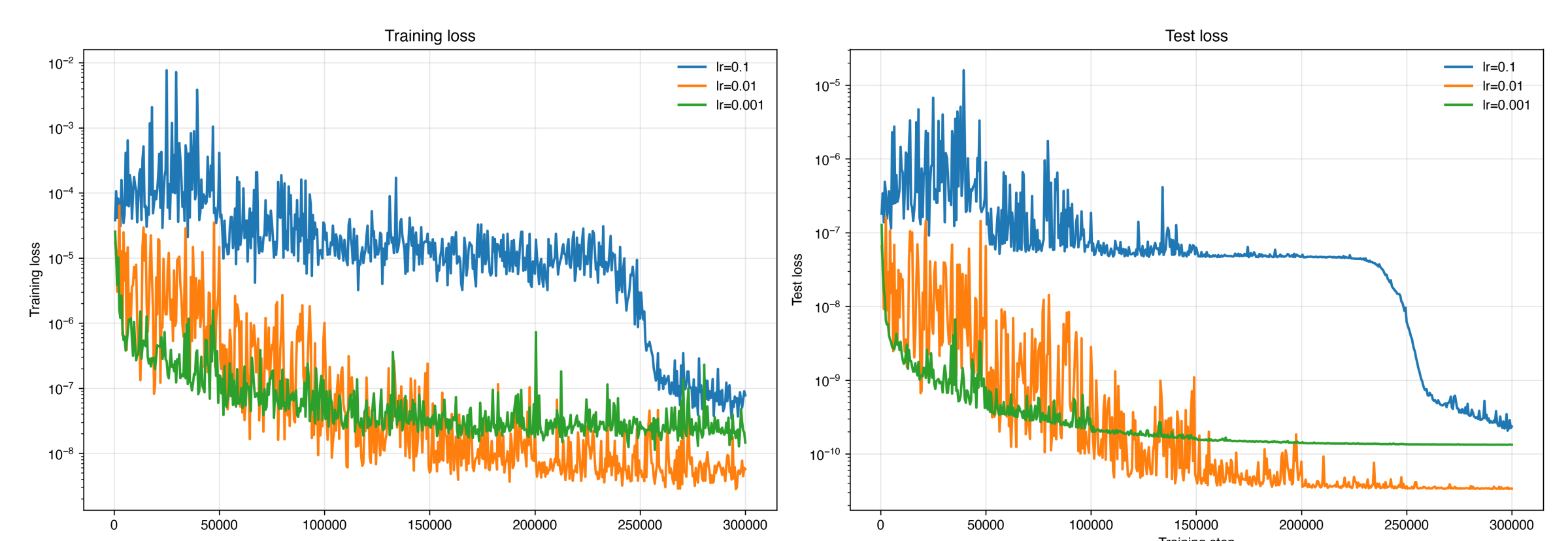
$$u_{k+1} = \mu_b \int \sigma(w^\top X_k + u_k) \lambda(dw), \quad W^i \sim \lambda, \quad \mathbb{E}[B^i] = \mu_b.$$

Fixed point analysis: If $(X_k, Z_k, Y_k, u_k) \sim \mu_k$, then there exists a **unique invariant measure** μ such that $\mu_k \xrightarrow{k \rightarrow \infty} \mu$.



Training Dynamics of a RNN

The limit of evolution truncated back-propagation through time is a **gradient flow** (see above), and a **kernel** over test functions emerges. [2]



References

- [1] Williams R, Peng J. An efficient gradient-based algorithm for on-line training of recurrent network trajectories. Neural Computation. 1990;2(4):490-501.
- [2] Spiliopoulos SCHLJSK. Kernel limit for a class of recurrent neural networks trained on ergodic data sequences. Annals of Applied Probability. 2026 June;36(3):2014-67.